Semantic Uniqueness and LIFO Lift Theory for Real-Time Pushdown Normalization

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Abstract

For the weighted balance languages

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}, \qquad \gcd(p,q) = 1,$$

we separate the unique semantic dynamics from the non-unique real-time pushdown realizations that lift those dynamics through literal LIFO memory. The machine-independent right-residual quotient is the integer translation system

$$r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q,$$

and the minimum-completion dynamics used in Write–Search–Cancel normalization is an isomorphic coordinatization of this quotient. We then show that a fixed canonical literal stack section is transition-closed under one-top real-time rewriting only in the trivial ratio $(p,q) = (1,1)$; every nontrivial ratio therefore forces noncanonical physical fibers in that policy class. This makes lift minimality representation- and morphism-relative rather than absolute.

The ratio 2:1 is developed as an exact case study. Three reachable eager-discharge sites generate exactly six admissible normalization schedules, characterized by the single constraint $ab = 0$. These six schedules give six pairwise non-isomorphic coordinate-reduced lifts in the stated coordinate category. The resulting schedule poset is not a lattice, yet all six schedules have the same canonical terminal normalization. A local interaction analysis explains the unique minimal conflict causally: enabling one site expands the first-hit observation fiber of another beyond its portability set, producing carry aliasing and a forced one-kernel-packet excess. This yields a directed causal schedule edge $A \to B$, invariant under site-preserving coordinate LIFO isomorphism but not determined by the semantic quotient. Finally, we introduce an obstruction-persistence spectrum across nested representation classes, separating presentation-level, fixed-policy, and broader-realization persistence. The central conclusion is that semantic uniqueness does not imply uniqueness of physical normalization structure, and that every claim of a forced pushdown obstruction is meaningful only relative to an explicit representation category.

1 Introduction

The language family $L_{p,q}$ is arithmetically simple: membership depends only on the signed residual

$$\rho(w) = qN_1(w) - pN_0(w).$$

Nevertheless, deterministic real-time pushdown implementations of the same residual dynamics may have very different physical stack behavior. A canonical literal representation may force temporary noncanonical configurations because a pushdown transition can rewrite only the visible stack top while preserving the untouched suffix. A different residual coordinate system can remove the same obstruction entirely.

This paper studies that gap between *semantic uniqueness* and *physical lift nonuniqueness*. It continues the structural Write–Search–Cancel (WSC) analysis of literal-demand normalization [1] and uses the configuration-conjugacy constructions of the companion geometry/resource paper [3] as a re-coordination witness. The centered real-time construction already supplies the machine family for every coprime positive p, q [2]; recognition of $L_{p,q}$ itself is therefore not the novelty target here. The questions are instead:

(i) what is the machine-independent semantic quotient of every correct deterministic real-time recognizer;

(ii) when a canonical semantic section can be realized directly by legal one-top transitions;

(iii) how minimality depends on the chosen morphism category;

(iv) how physically different normalization schedules can remain semantically and terminally equivalent;

(v) how schedule conflicts can be detected locally and assigned causal direction; and

(vi) which physical obstructions survive changes of coordinates or policy.

The main structural message is

$$\boxed{\text{unique semantic core does not imply unique physical normalization structure.}}$$

A second methodological message is equally important:

$$\boxed{\text{“minimal”, “forced”, and “obstruction” require a representation category.}}$$

Scope. The general semantic and locality theorems are stated for all coprime $p, q \geq 1$. The exact schedule classification, causal graph, and bounded terminal-normalization results are proved for the selected three-site 2:1 literal-policy family. We do not claim unrestricted DPDA minimality, a general confluence theorem, downward closure for arbitrary schedule systems, or a complete classification of all physical lifts.

Related work and novelty boundary. Language residuals and congruence quotients are classical semantic tools. For visibly pushdown languages, Alur, Kumar, Madhusudan, and Viswanathan developed congruence-based canonical automata and showed that, unlike the DFA situation, canonicality does not in general coincide with unique size-minimal realization [4]. Gauwin, Muscholl, and Raskin later proved NP-completeness of visibly pushdown automaton minimization and emphasize that minimal VPAs need not be unique [5]. At the unrestricted DPDA level, language equivalence is decidable [6, 7]; this is a behavioral question and does not supply a canonical physical stack presentation. The present contribution is therefore stated narrowly: for the weighted-balance normalization family we identify the machine-independent residual quotient, prove a fixed-policy one-top branching obstruction, and develop representation-relative schedule, causality, and obstruction-persistence notions. We do not claim a general DPDA minimization theorem or a new form of classical Church–Rosser confluence.

2 Semantic preliminaries

Fix coprime integers $p, q \geq 1$ and let

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

Define

$$\rho(w) = qN_1(w) - pN_0(w).$$

For a suffix with x zeros and y ones, completion of a prefix of residual r requires

$$px - qy = r.$$

The primitive zero-residual kernel packet is

$$K = (q, p).$$

The WSC minimum completion map will be denoted

$$\mu : \mathbb{Z} \to M_{p,q} \subseteq \mathbb{N}_0^2.$$

Equivalently, if a prefix w has $a = N_0(w)$ and $b = N_1(w)$, then

$$\kappa(w) = \max\left(\left\lceil \frac{a}{q} \right\rceil, \left\lceil \frac{b}{p} \right\rceil\right),$$

$$\mu(w) = \big(q\kappa(w) - a,\ p\kappa(w) - b\big).$$

The canonical literal orientation used throughout the literal-policy sections is

$$\eta(x, y) = 1^y 0^x Z_0,$$

with the stack top written on the left.

Definition 2.1 (Residual semantic system)**.** The canonical residual system is

$$\mathcal{R}_{p,q} = \big(\mathbb{Z}, U_0, U_1, 0\big),$$

where

$$U_0(r) = r - p, \qquad U_1(r) = r + q,$$

and residual 0 is the accepting semantic state.

3 Semantic uniqueness

Theorem 3.1 (Right-residual classification)**.** *For all* $u, v \in \{0,1\}^*$,

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

Moreover, every integer residual is attained by some word.

Proof. If $\rho(u) = \rho(v)$, then for every suffix z,

$$\rho(uz) = \rho(u) + \rho(z) = \rho(v) + \rho(z) = \rho(vz),$$

so $uz \in L_{p,q}$ iff $vz \in L_{p,q}$.

Conversely suppose $\rho(u) \neq \rho(v)$. Write $r = \rho(u)$ and $s = \rho(v)$. Because $\gcd(p, q) = 1$, there exist integers x_0, y_0 with $px_0 - qy_0 = r$. Adding a sufficiently large multiple of (q, p) makes both coordinates nonnegative, so some suffix z satisfies $px - qy = r$. Then $\rho(uz) = 0$, whereas $\rho(vz) = s - r \neq 0$. Hence the right residual languages differ.

For attainability, Bezout gives integers m, n with $qm - pn = 1$. Multiplying by any $r \in \mathbb{Z}$ and then adding a sufficiently large balanced kernel shift yields nonnegative symbol counts realizing residual r. $\square$

Corollary 3.2 (Canonical behavioral core)**.** *The reachable right-residual classes of $L_{p,q}$ are canonically identified with $\mathbb{Z}$, and the induced labeled transition system is $\mathcal{R}_{p,q}$.*

Definition 3.3 (Input-boundary configuration)**.** For a deterministic recognizer with initialization or finalization ε-moves, an input-boundary configuration means the deterministic closed configuration immediately before or after consumption of an ordinary input symbol. In an ε-free core this is simply the ordinary reached configuration.

Theorem 3.4 (Universal semantic quotient)**.** *Let $\mathcal{A}$ be any correct deterministic real-time recognizer for $L_{p,q}$, interpreted on reachable input-boundary configurations. If c_u is the configuration reached after prefix u, then*

$$\pi_{\mathcal{A}}(c_u) = \rho(u)$$

is well defined, surjective, and transition preserving:

$$\pi_{\mathcal{A}}(\delta_0(c)) = \pi_{\mathcal{A}}(c) - p, \qquad \pi_{\mathcal{A}}(\delta_1(c)) = \pi_{\mathcal{A}}(c) + q.$$

It is the unique such quotient map sending the initial configuration to residual 0.

Proof. Every prefix is extendable to a word of $L_{p,q}$ by Theorem 3.1, so a correct recognizer cannot become permanently blocked after an ordinary prefix; the input-boundary configuration c_u is therefore defined for every u. If $c_u = c_v$, determinism implies that u and v have identical continuation languages. By Theorem 3.1, $\rho(u) = \rho(v)$, so $\pi_{\mathcal{A}}$ is well defined. Surjectivity follows from attainability of all residuals. The transition equations follow from

$$\rho(u0) = \rho(u) - p, \qquad \rho(u1) = \rho(u) + q.$$

Uniqueness follows inductively from the initial value 0 and the two labeled increments. □

Theorem 3.5 (WSC semantic core equals the residual core)**.** *The map*

$$J : \mathbb{Z} \to M_{p,q}, \qquad J(r) = \mu(r),$$

is a bijection and intertwines the residual updates with the WSC minimum-demand updates. Hence

$$\boxed{\mathcal{R}_{p,q} \cong M_{p,q}.}$$

Proof. The kernel-fiber theorem and minimum-nonnegative representative theorem of WSC give a unique $\mu(r)$ for every r, so J is bijective. If $e_0 = (1,0)$ and $e_1 = (0,1)$, WSC gives

$$\mu' = \mu - e_a + \chi K, \qquad \chi \in \{0,1\},$$

which is exactly the unique minimum representative of residual $r - p$ on input 0 and $r + q$ on input 1. Thus J conjugates the labeled dynamics. □

4 Sectioned lifts and morphism-relative minimality

Definition 4.1 (Exact sectioned lift)**.** Let $\mathcal{S} = (M, \{U_a\}_{a\in\Sigma})$ be a semantic system. An exact sectioned lift is a tuple

$$\mathcal{L} = (C^{\text{reach}}, \delta, \Theta, \sigma)$$

where C^{reach} is the run-reachable input-boundary configuration set, δ_a are deterministic physical transitions on reachable configurations, $\Theta : C^{\text{reach}} \to M$ is a surjective semantic projection, and

$$\sigma : M \to C^{\text{reach}}$$

is a *reachable section* satisfying

$$\Theta \circ \sigma = \mathrm{id}_M, \qquad \Theta(\delta_a(c)) = U_a(\Theta(c)).$$

When a concrete pushdown syntax is needed we may also refer to a larger ambient set $C^{\mathrm{amb}} \supseteq C^{\mathrm{reach}}$ of syntactically valid configurations, but all minimality, fiber, and conflict claims in this paper are about C^{reach} unless explicitly stated otherwise.

Definition 4.2 (Section-preserving morphism)**.** A morphism $F : \mathcal{L}_1 \to \mathcal{L}_2$ between exact sectioned lifts satisfies

$$\Theta_2 F = \Theta_1, \qquad F\delta_a^{(1)} = \delta_a^{(2)} F, \qquad F\sigma_1 = \sigma_2.$$

Definition 4.3 (Coordinate LIFO morphism)**.** For pushdown lifts with control sets Q_i and stack alphabets Γ_i, a coordinate LIFO morphism is determined by maps

$$f : Q_1 \to Q_2, \qquad g : \Gamma_1 \to \Gamma_2,$$

with bottom marker preserved and g extended letterwise to $g^* : \Gamma_1^* \to \Gamma_2^*$. On configurations,

$$F(q, S) = \big(f(q), g^*(S)\big).$$

It must also be a section-preserving transition morphism and preserve acceptance. A coordinate quotient is such a morphism with f and g surjective.

Definition 4.4 (Coordinate-reduced lift)**.** A lift in a fixed physical policy class is coordinate-reduced if every surjective coordinate LIFO morphism from it to another lift in the same policy class is an isomorphism.

Proposition 4.5 (Minimality is category-relative)**.** *There is no representation-independent notion of "the minimal lift" determined solely by the semantic quotient. Any nontrivial minimality statement requires an admissible physical policy class and a morphism category.*

Proof. Every exact lift admits the abstract semantic quotient obtained by identifying configurations with equal semantic projection. If arbitrary behavioral quotients are admitted, all physical lifts collapse to the same semantic system. In contrast, coordinate LIFO quotients preserve state and stack-symbol coordinates letterwise and may fail to collapse physically distinct lifts. Hence the set of admissible reductions depends on the chosen morphism category. □

5 One-top locality and forced fiber branching

A standard PDA transition rewrites only the visible stack top. Write a source stack as XT, where X is the top symbol and T is the untouched tail. The following one-top criterion is recalled from WSC [1] for self-containment.

Lemma 5.1 (Top-rewrite suffix criterion (WSC, restated))**.** *From a source stack XT, a desired target stack S^* is realizable in one top rewrite iff there exists a word γ such that*

$$S^* = \gamma T.$$

Equivalently, the untouched tail T must be a suffix of S^*.

Proof. A single PDA transition replacing X by γ produces exactly γT, and no other part of the stack can change. □

Theorem 5.2 (Uniform one-top rewrite criterion)**.** *Consider source configurations*

$$c_i = (s, XT_i)$$

sharing the same control state, input symbol, and visible top symbol. One deterministic top rule realizes desired targets (s_i^*, S_i) *for all* i *iff there exist one control state* s' *and one replacement word* γ *such that*

$$s_i^* = s', \qquad S_i = \gamma T_i$$

for every i.

Proof. A deterministic PDA cell indexed by (s, a, X) has one fixed target control state and one fixed replacement word. The conclusion is therefore necessary and sufficient. □

Theorem 5.3 (Canonical-section closure dichotomy)**.** *Fix the WSC canonical literal section*

$$\sigma(x, y) = (C, 1^y 0^x Z_0).$$

This section is transition-closed under exact deterministic one-top real-time rewriting iff

$$\boxed{(p, q) = (1, 1).}$$

Proof. For $(p, q) = (1, 1)$, the minimum-demand state has at most one nonzero coordinate: otherwise $(x-1, y-1)$ is a shorter nonnegative representative of the same residual. Hence canonical stacks are unary and direct top cancellation or one-symbol carry preserves the canonical form.

Now assume $(p, q) \neq (1, 1)$. The canonical semantic states

$$m_A = (0, 1), \qquad m_B = (1, 1)$$

exist; each is a minimum representative because subtracting one kernel packet would make at least one coordinate negative. Their canonical stacks are respectively $1Z_0$ and $10Z_0$, so on input 0 both present the same local observation $(C, 0, 1)$ at the transition cell, but the required semantic actions differ: at m_A the zero-coordinate is exhausted and a kernel carry is required, while at m_B the zero demand is directly cancellable only below the visible 1. The two canonical targets cannot both be obtained by one fixed top rewrite. Equivalently, for the buried-cancellation source $10Z_0$, the untouched suffix $0Z_0$ is not a suffix of the direct canonical cancellation target. Thus the canonical section is not transition-closed. □

Corollary 5.4 (Essential physical fiber branching)**.** *For every nontrivial coprime ratio* $(p, q) \neq (1, 1)$*, any exact one-top real-time lift in the fixed WSC canonical literal policy that contains the canonical section has a reachable semantic fiber containing a noncanonical physical representative in addition to its canonical section point.*

Proof. By Theorem 5.3, some reachable canonical configuration has a legal exact successor that cannot equal the canonical target. Exactness places that successor in the same semantic fiber as the canonical target, while the two configurations are physically distinct. □

Corollary 5.5 (Quotient-protected branching)**.** *Within the fixed canonical literal one-top policy, no admissible section-preserving quotient can identify every locality-forced noncanonical successor with its canonical section target while preserving the legal one-top transition structure.*

Proof. Such an identification would turn the corresponding physical transition into a direct canonical-section transition, contradicting Theorem 5.3 and the suffix criterion. □

6 The 2:1 eager/lazy lifts

Set $(p, q) = (2, 1)$ and $K = (1, 2)$. We use the three core control states C, L, R of the motivating literal-demand architecture. The centered realization keeps certain finite-control debt in a side state, whereas the historical realization materializes the same debt immediately on two reachable right-state matching families [2, 1].

For example,

$$(R, 0) \xrightarrow{0}_{\text{lazy}} (R, \varepsilon), \qquad (R, 0) \xrightarrow{0}_{\text{eager}} (C, 111),$$

and for $k \geq 1$,

$$(R, 1^k) \xrightarrow{1}_{\text{lazy}} (R, 1^{k-1}), \qquad (R, 1^k) \xrightarrow{1}_{\text{eager}} (C, 1^{k+2}).$$

The two successors in each pair have the same semantic projection.

Theorem 6.1 (Eager–lazy section incomparability)**.** *The centered lazy lift and the historical eager lift are not related by a section-preserving transition morphism in either direction.*

Proof. Consider the common reachable configurations

$$c_2 = (C, 11), \qquad c_1 = (C, 1),$$

with

$$c_2 \xrightarrow{0} r_1 = (R, 1), \qquad c_1 \xrightarrow{0} r_0 = (R, \varepsilon).$$

In the lazy lift,

$$r_1 \xrightarrow{1} r_0,$$

whereas in the eager lift,

$$r_1 \xrightarrow{1} c_3 = (C, 111).$$

A section-preserving morphism fixes the canonical section configurations and preserves transitions. Mapping r_1 compatibly with the shared predecessor forces the same image in both systems, but its input-1 successor would then have to be simultaneously r_0 and c_3, which are distinct. The reverse direction yields the same contradiction. □

Proposition 6.2 (Coordinate reducedness in the three-state/two-work-symbol presentation)**.** *Within the fixed* 2:1 *coordinate category of Theorem 4.3, the eager and lazy lifts are coordinate-reduced.*

Proof. Reachable witnesses $(C, \varepsilon), (L, \varepsilon), (R, \varepsilon)$ carry three distinct residuals $0, 3, -3$, so any exact coordinate quotient must keep the three control states distinct. Reachable witnesses $(C, 0)$ and $(C, 1)$ carry distinct residuals and force the two work symbols to remain distinct. Thus every surjective coordinate map is bijective on the relevant control and stack-symbol coordinates. Because the reachable systems are generated from their section/initial configurations and the morphism commutes with every labeled transition, this coordinate bijection induces a bijection on reachable configuration sets; its inverse also preserves the labeled transition structure and the section. Hence the quotient morphism is an isomorphism. □

Corollary 6.3 (Morphism-relative nonuniqueness)**.** *The same semantic quotient admits multiple coordinate-reduced physical lifts that are not section-preserving transition-isomorphic.*

7 Normalization schedules in the 2:1 family

Throughout this case study the binary-input core starts at (C, ε), and the fixed acceptance wrapper accepts exactly when end-of-input normalization reaches (C, ε). In particular, a residual-zero noncanonical center word is not silently erased by the wrapper. This convention is the one used by the centered/historical literal-demand constructions cited above.

We now isolate three reachable optional eager-discharge sites. Each site replaces a lazy matching discharge by immediate canonical materialization of the side-state debt.

Definition 7.1 (Three schedule sites)**.** Let

$$A: \quad (L, 0T) \xrightarrow{0} \begin{cases} (L, T), & a = 0, \\ (C, 100T), & a = 1, \end{cases}$$

$$B: \quad (R, 0T) \xrightarrow{0} \begin{cases} (R, T), & b = 0, \\ (C, 111T), & b = 1, \end{cases}$$

$$C: \quad (R, 1T) \xrightarrow{1} \begin{cases} (R, T), & c = 0, \\ (C, 111T), & c = 1. \end{cases}$$

A schedule is a bit vector $(a, b, c) \in \{0, 1\}^3$.

Definition 7.2 (Finite schedule system)**.** For a finite set of optional local sites E, let $\mathcal{L}_H$ denote the hybrid lift obtained by enabling exactly $H \subseteq E$. Define

$$\mathrm{Adm}(E) = \big\{H \subseteq E : \mathcal{L}_H \text{ is deterministic, exact, in policy,} \\ \text{and recognizes the target language}\big\}.$$

A minimal schedule conflict is a nonadmissible $K \subseteq E$ whose every proper subset is admissible.

Theorem 7.3 (Exact three-site schedule classification)**.** *For the selected* 2:1 *three-site family,*

$$\boxed{\mathrm{Adm}_{2,1} = \{(a, b, c) \in \{0, 1\}^3 : ab = 0\}.}$$

Equivalently, the six admissible schedules are

$$000, 001, 010, 011, 100, 101,$$

and the only minimal conflict is

$$\boxed{\mathrm{Crit}_{2,1} = \{\{A, B\}\}.}$$

Proof. First suppose $a = 0$. A direct induction on the transition table yields the original reachable-store invariant

$$C: \ 0^* \cup 1^* \cup \{10\}, \qquad L: \ 0^*, \qquad R: \ 1^* \cup \{0\}.$$

The optional B and C rules return directly to canonical center words, preserve this family, and satisfy the same residual update, so exactness is retained. Using the state potentials $\Phi(C) = 0$, $\Phi(L) = 3$, $\Phi(R) = -3$ and stack weight $D(S) = 2N_0(S) - N_1(S)$, none of the nonempty configurations in these families has residual zero; hence recognition is correct for all four schedules with $a = 0$.

Now suppose $b = 0$. A direct induction on the transition table yields the invariant family

$$C: \ \varepsilon \cup 0^+ \cup 1^+ \cup 10^+, \qquad L: \ 0^*, \qquad R: \ 0^* \cup 1^*.$$

Within this family, enabling A and/or C preserves exactness. Again using $D(S) + \Phi(s_{\mathrm{ctl}})$, the side families cannot have residual zero, unary nonempty center words cannot have residual zero, and a center word 10^n has residual $2n - 1 \neq 0$. Thus only (C, ε) has residual zero. Hence every schedule with $ab = 0$ is admissible.

It remains to exclude $a = b = 1$. Starting from (C, ε), the prefix 1111100 reaches $(R, 00)$ without using a B-transition: the crucial step is the eager A discharge

$$(L, 0) \xrightarrow{0} (C, 100),$$

followed by an ordinary zero transition to $(R, 00)$. Enabling B then gives

$$(R, 00) \xrightarrow{0} (C, 1110).$$

The resulting residual is -1, whose canonical literal stack is 1, so the center stack 1110 contains one excess kernel packet. Appending input 1 yields

$$(C, 1110) \xrightarrow{1} (C, 110),$$

at residual zero but away from the canonical accepting configuration. Thus both schedules 110 and 111 fail. This proves the classification. □

Corollary 7.4 (Schedule geometry). *The admissible schedule poset under coordinatewise lazy-to-eager order is not a lattice. Its maximal schedules are*

$$\boxed{\{A, C\} \quad \textit{and} \quad \{B, C\}.}$$

Proof. The admissible sets are exactly those not containing $\{A, B\}$. Thus $\{A, C\}$ and $\{B, C\}$ have no admissible common upper bound, so the poset has no join for this pair. □

Proposition 7.5 (Six coordinate-reduced schedule lifts). *Within the fixed coordinate category, the six admissible schedules give six pairwise non-isomorphic coordinate-reduced lifts.*

Proof. The witnesses used in Theorem 6.2 force every coordinate isomorphism to fix C, L, R and the work symbols $0, 1$. Each of the three schedule sites is reachable. Therefore two distinct schedule bit vectors differ on a reachable transition that cannot be renamed away. The same coordinate witnesses prohibit a nontrivial surjective coordinate quotient, and the reachable-generation argument from Theorem 6.2 makes every remaining surjective coordinate morphism an isomorphism. □

8 Terminal normalization equivalence

Definition 8.1 (Terminal normalizer). Fix the administrative end-of-input normalizer $\mathfrak{N}$ by

$$\begin{aligned}
(L, \varepsilon) &\mapsto (C, 100), \\
(L, 0T) &\mapsto (C, 1000T), \\
(R, \varepsilon) &\mapsto (C, 111), \\
(R, 1T) &\mapsto (C, 1111T), \\
(R, 0T) &\mapsto (C, 1T),
\end{aligned}$$

while every reachable canonical center configuration in the admissible family is fixed. The map is only asserted on the reachable families used in the theorem below; no rule is added that erases an arbitrary noncanonical center word. This is a normalization map, not an ordinary ε-execution convention.

Definition 8.2 (Terminal coherence)**.** A schedule α is terminally coherent relative to $\mathfrak{N}$ if for every prefix w,

$$\mathfrak{N}(c_\alpha(w)) = \sigma\big(\Theta(c_\alpha(w))\big).$$

Two schedules are terminal-normalization equivalent if their normalized endpoints agree after every prefix.

Theorem 8.3 (Six-schedule terminal coherence)**.** *All six admissible schedules of Theorem 7.3 are terminally coherent relative to $\mathfrak{N}$. Hence they are pairwise terminal-normalization equivalent.*

Proof. For schedules with $a = 0$, the reachable stores are the original centered/historical families; the five clauses of $\mathfrak{N}$ are exactly their deterministic canonical completion rules. For schedules with $b = 0$ and $a = 1$, the only new right-state family is $R0^n$. The clause

$$(R, 0T) \mapsto (C, 1T)$$

sends $R0^n$ to $C10^{n-1}$, the canonical literal stack for the same residual. The C-site choice changes only whether right-state 1-debt is discharged before or at terminal normalization, so both choices have the same normalized endpoint. Thus every reachable configuration in each of the six schedules normalizes to the unique canonical section representative of its semantic state. □

Corollary 8.4 (Reduced non-isomorphism with common normalized behavior)**.** *There exist coordinate-reduced, pairwise non-isomorphic lifts with identical terminal-normalization behavior on every prefix.*

Remark 8.5. We call this *terminal schedule confluence* only in the explicitly normalizer-relative sense above. No claim of classical Church–Rosser confluence is made.

9 Defect channels and local interaction

For a literal stack word S, let $P(c) = (N_0(S), N_1(S))$ be its physical count vector and let $\Theta(c) = \mu(\rho(c))$ be the canonical demand vector. Following WSC, define the quantity defect

$$\boxed{Q(c) = \Theta(c) - P(c).}$$

This definition is purely count-level: finite-control debt appears through the resulting defect coordinates rather than by altering $P(c)$. Define the order inversion count

$$\Omega(S) = \#\{(i, j) : i < j,\ S_i = 0,\ S_j = 1\}.$$

Let $\xi(c)$ be a control-canonicality indicator, equal to 0 exactly on the chosen canonical section state for that semantic representative.

Definition 9.1 (Canonicality defect signature)**.**

$$\mathfrak{D}(c) = \big(Q(c), \Omega(c), \xi(c)\big).$$

The three-channel canonicality principle is inherited from WSC [1]; we restate it here in the signature notation above.

Proposition 9.2 (Three-channel canonicality (WSC, restated))**.** *In the fixed canonical literal policy,*

$$\boxed{\mathfrak{D}(c) = 0 \iff c = \sigma(\Theta(c)).}$$

Proof. Zero quantity defect gives the canonical Parikh vector, zero order defect puts all 1 symbols before all 0 symbols in the fixed orientation, and zero control defect selects the canonical control state. Together these determine exactly $\sigma(\Theta(c))$. Conversely the canonical section has all three defects zero. □

Proposition 9.3 (Configuration-level separation of the three defect channels)**.** *At the level of exact semantic representations allowed in this paper, none of the quantity, order, and control defect conditions is implied by the other two: each channel has a witness with the other two equal to zero.*

Proof. Quantity-only failure is witnessed by the $A + B$ kernel-excess configuration $(C, 1110)$ against canonical $(C, 1)$, for which $Q = -K$ while order and control are canonical. Order-only failure is witnessed by the WSC carry-order family, which preserves the canonical Parikh vector while creating positive inversion count. Control-only failure is witnessed explicitly by a two-sheet decoration of the exact 2:1 literal lift: update the auxiliary sheet bit on every consumed symbol while leaving the base stack dynamics unchanged. The balanced word 011 reaches the base canonical configuration (C, ε) after three symbols, so the decorated endpoint has canonical stack counts and order but the noncanonical sheet bit. The last witness belongs to an enlarged exact-lift presentation rather than to one fixed three-state control table. □

9.1 Materialization and carry aliasing

In WSC phase-coherent count coordinates, write the defect as

$$Q = sh + wK.$$

If a physical transition changes stack length by $\Delta|P|$, its materialized kernel amount is

$$M = \frac{\Delta|P| + 1}{p + q},$$

and the backlog law is

$$\boxed{\Delta w = \chi - M.}$$

Here $\chi \in \{0, 1\}$ is the semantic carry determined by the full semantic context.

Theorem 9.4 (Local carry visibility criterion)**.** *Within a phase-coherent WSC count-shadow realization, fix a local PDA observation cell $o = (s, a, X)$ and restrict to zero-backlog source configurations in that cell. A single fixed local rule can be perfectly eager and backlog-neutral throughout the observation fiber only if the semantic carry is constant on that fiber:*

$$\boxed{\chi(c, a) = \bar{\chi}(o).}$$

If the fiber contains contexts with both $\chi = 0$ and $\chi = 1$, no fixed materialization amount M can satisfy $M = \chi$ everywhere.

Proof. A deterministic cell has one fixed stack replacement and hence one fixed M. At zero backlog, perfect eager neutrality requires $0 = \Delta w = \chi - M$. Therefore M must equal every carry value appearing in the observation fiber. □

Definition 9.5 (Portability set)**.** For a local eager rule r at observation o, define

$$\mathrm{Port}(r) = \{c \in \mathrm{Reach} : \text{ applying } r \text{ to } c \text{ gives } \sigma(U_a(\Theta(c)))\}.$$

For a schedule α, let F_o^α be the reachable observation fiber at o.

Proposition 9.6 (Eager-rule portability criterion)**.** *The eager rule r is directly canonical on every reachable occurrence under schedule α iff*

$$\boxed{F_o^\alpha \subseteq \mathrm{Port}(r).}$$

9.2 The $A \to B$ interaction

Lemma 9.7 (Schedule-induced fiber expansion). *With A enabled, for every $m \geq 2$ the configuration $(R, 0^m)$ is reachable before any use of the B transition cell.*

Proof. For every $n \geq 1$, the ordinary run reaches $(L, 0^n)$ after the prefix 1^{2n+3}. Enabling A and reading 0 gives

$$(L, 0^n) \xrightarrow{0} (C, 10^{n+1}),$$

and one further ordinary zero transition gives

$$(C, 10^{n+1}) \xrightarrow{0} (R, 0^{n+1}).$$

Taking $m = n + 1$ proves the claim. □

Theorem 9.8 (Schedule-induced direct-canonicality obstruction). *For every $m \geq 2$, no one-top real-time transition on input 0 can send $(R, 0^m)$ directly to its canonical successor.*

Proof. After reading 0, the canonical target stack is 10^{m-2}. A one-top rewrite of source 0^m must leave the untouched suffix 0^{m-1} intact. But 0^{m-1} is not a suffix of 10^{m-2}. Apply Theorem 5.1. □

Theorem 9.9 (Forced kernel-excess return). *Let $m \geq 2$. Any residual-preserving one-top transition*

$$(R, 0^m) \xrightarrow{0} (C, \gamma 0^{m-1})$$

that returns directly to the center produces at least one excess kernel packet relative to the canonical target. If

$$P(\gamma) = (t, 3 + 2t), \qquad t \geq 0,$$

then the excess is exactly

$$\boxed{(t + 1)K.}$$

The historical replacement $\gamma = 111$ realizes the minimum excess K.

Proof. Residual preservation forces the replacement word to have weight -3, hence

$$2N_0(\gamma) - N_1(\gamma) = -3.$$

All nonnegative solutions are $(t, 3 + 2t)$. The resulting stack has Parikh vector

$$(m - 1 + t, 3 + 2t),$$

whereas the canonical target 10^{m-2} has vector $(m - 2, 1)$. Their difference is

$$(t + 1, 2t + 2) = (t + 1)K.$$

□

Corollary 9.10 (Three-way diagnosis of the $A \to B$ conflict). *On the A-induced family $(R, 0^m)$, $m \geq 2$, the same obstruction admits three equivalent diagnoses within this family:*

(i) *failure of the one-top suffix condition for direct canonicalization;*

(ii) *carry aliasing with fixed eager materialization $M = 1$ but semantic carry $\chi = 0$;*

(iii) *forced positive kernel excess in every one-top center-return repair.*

10 Directed causal schedule structure

Definition 10.1 (First-hit exposure fiber)**.** Let b be a schedule site and let H be a set of other enabled sites. The first-hit exposure fiber $\mathcal{E}_b(H)$ consists of configurations reachable by paths that use no b-transition and from which the next designated input would trigger the local cell of b. Define the unsafe first-hit fiber

$$\mathcal{U}_b(H) = \mathcal{E}_b(H) \setminus \mathrm{Port}(b).$$

Definition 10.2 (Primary causal dependency)**.** For $H \subseteq E \setminus \{b\}$, write

$$H \rightsquigarrow b$$

if

(i) H is admissible;

(ii) $\mathcal{U}_b(H) \neq \varnothing$; and

(iii) $\mathcal{U}_b(H') = \varnothing$ for every proper subset $H' \subsetneq H$.

Theorem 10.3 (Exact primary dependency classification for 2:1)**.** *For the selected three-site family,*

$$\boxed{\mathcal{D}^{\mathrm{prim}}_{2,1} = \{\{A\} \rightsquigarrow B\}.}$$

Thus the directed causal schedule graph is

$$\boxed{A \longrightarrow B \qquad \sqcup \qquad C.}$$

Proof. By Theorem 9.7, enabling A exposes $(R, 00)$ to the B cell without any prior B use. By Theorem 9.8, this configuration is outside $\mathrm{Port}(B)$, while the baseline right-top-zero source $(R, 0)$ is portable. Hence $A \rightsquigarrow B$.

The A rule is uniformly portable on every reachable first-hit A source: the reachable left store is always 0^n, and replacing the visible 0 by 100 sends $(L, 0^n)$ to the canonical target $C10^{n+1}$ for every $n \geq 1$. Hence neither B nor C, alone or together when admissible, can expose an unsafe first hit for A. Similarly, every first-hit C source in an admissible cause schedule is of the form $(R, 1^k)$; replacing the top 1 by 111 produces the canonical target $C1^{k+2}$ for all $k \geq 1$. Finally, enabling C returns such sources to canonical unary center words and does not enlarge the top-zero B observation fiber beyond the baseline portable source $(R, 0)$ unless A is enabled. Therefore C creates no primary unsafe exposure for A or B, B creates none for A or C, and there are no other primary dependencies. □

Corollary 10.4 (Directed cause recovers the minimal conflict)**.** *For this model, forgetting the direction of the unique primary causal edge recovers the unique minimal schedule conflict:*

$$A \rightsquigarrow B \quad \rightsquigarrow \quad \{A, B\} \in \mathrm{Crit}_{2,1}\,.$$

Definition 10.5 (Channel-labelled causal edge)**.** If a primary dependency produces a first post-normalization defect with support in a channel set $\Lambda \subseteq \{\mathsf{Q}, \mathsf{O}, \mathsf{C}\}$, label the causal hyperedge by Λ and, when available, by the exact defect charge.

Proposition 10.6 (2:1 causal certificate)**.** *The unique primary edge has quantity-channel label*

$$\boxed{A \xrightarrow{\mathsf{Q}:-K} B.}$$

A first-hit certificate is

$$1111100 \rightsquigarrow (R, 00),$$

followed by

$$(R, 00) \xrightarrow[B]{0} (C, 1110),$$

whose canonical target is $(C, 1)$*. Appending* 1 *yields the residual-zero noncanonical configuration* $(C, 110)$*.*

11 Coordinate invariance and semantic non-determination

Definition 11.1 (Schedule presentation isomorphism)**.** A site-preserving coordinate isomorphism between schedule presentations consists of a coordinate LIFO isomorphism F together with a bijection φ between optional sites, carrying each local observation cell and its replacement word letterwise to the corresponding site in the target presentation.

Theorem 11.2 (Causal hypergraph invariance)**.** *Under a schedule presentation isomorphism* (F, φ)*,*

$$F(\mathcal{E}_b(H)) = \mathcal{E}_{\varphi(b)}(\varphi(H)),$$
$$F(\mathrm{Port}(b)) = \mathrm{Port}(\varphi(b)),$$

and therefore

$$\boxed{H \rightsquigarrow b \iff \varphi(H) \rightsquigarrow \varphi(b).}$$

Thus the directed causal schedule hypergraph is a coordinate-structural invariant of the presentation.

Proof. Coordinate isomorphism preserves labeled reachability, local top observations, the property of avoiding a designated site before first hit, exact semantic targets, and canonical section points. Hence it carries exposure fibers and portability sets bijectively. Minimality of the causing set is preserved by the site bijection. □

Definition 11.3 (Causal schedule signature)**.** The causal schedule signature of a presentation is the isomorphism class

$$\mathrm{CSig}(\mathfrak{P}) = [\mathcal{H}^{\rightarrow}(\mathfrak{P})]$$

of its directed causal hypergraph.

Theorem 11.4 (Semantic non-determination of causal structure)**.** *The canonical semantic quotient does not determine the causal schedule signature.*

Proof. The literal 2:1 presentation above has the nontrivial causal edge $A \rightsquigarrow B$. By contrast, the companion raw residual-conjugacy construction [3] gives an ε-free right-endmarker DPDA with an injective encoding $\zeta : \mathbb{Z} \to Q \times \Gamma^*$ satisfying

$$\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)$$

for every residual r. Its ordinary reachable binary-input configuration graph is already label-isomorphic to $\mathcal{R}_{p,q}$, so the literal A, B, C debt-discharge schedule structure is absent. Both realize the same semantic quotient, but their physical causal schedule structures differ. □

Corollary 11.5 (Coordinate/semantic separation principle)**.** *A physical obstruction may be intrinsic to a coordinate-isomorphism class without being intrinsic to the language or to its canonical semantic quotient.*

12 Representation obstruction spectrum

Definition 12.1 (Nested representation classes)**.** For a fixed semantic system, consider nested lift classes

$$\mathfrak{C}_0 \subseteq \mathfrak{C}_1 \subseteq \mathfrak{C}_2,$$

where $\mathfrak{C}_0$ is one coordinate-isomorphism class, $\mathfrak{C}_1$ is a fixed physical policy class, and $\mathfrak{C}_2$ is a specified broader realization class of exact deterministic real-time lifts. The third level is therefore relative to the chosen broad realization universe; it should not be read as absolute semantic necessity unless $\mathfrak{C}_2$ has explicitly been chosen to contain every admissible realization under consideration. An obstruction property $\mathcal{O}$ is $\mathfrak{C}_i$-essential if every lift in $\mathfrak{C}_i$ has $\mathcal{O}$, and $\mathfrak{C}_i$-removable if some lift in $\mathfrak{C}_i$ does not.

Theorem 12.2 (Obstruction monotonicity)**.** *If*

$$\mathfrak{C}_0 \subseteq \mathfrak{C}_1 \subseteq \mathfrak{C}_2,$$

then

$$\boxed{\mathrm{Ess}(\mathfrak{C}_2) \subseteq \mathrm{Ess}(\mathfrak{C}_1) \subseteq \mathrm{Ess}(\mathfrak{C}_0).}$$

Proof. An obstruction holding for every member of a larger class necessarily holds for every member of each subclass. □

Definition 12.3 (Persistence profile)**.** For an obstruction $\mathcal{O}$, define

$$\Pi(\mathcal{O}) = (e_0, e_1, e_2), \qquad e_i = 1 \iff \mathcal{O} \text{ is } \mathfrak{C}_i\text{-essential}.$$

By Theorem 12.2, only

$$111, \quad 110, \quad 100, \quad 000$$

can occur for a fully resolved three-level profile.

Theorem 12.4 (Policy-essential but broad-class removable branching)**.** *Fix nested classes as in Theorem 12.1 with $\mathfrak{C}_1$ equal to the canonical literal one-top policy and with $\mathfrak{C}_2$ chosen to contain the raw residual-conjugacy re-coordination witness of [3]. For every nontrivial ratio $(p, q) \neq (1, 1)$, nontrivial semantic-fiber branching is $\mathfrak{C}_1$-essential but $\mathfrak{C}_2$-removable. Consequently, for any coordinate-isomorphism class $\mathfrak{C}_0 \subseteq \mathfrak{C}_1$,*

$$\boxed{\Pi(\mathcal{O}_{\mathrm{branch}}) = 110.}$$

Proof. Policy-essentiality is Theorem 5.4; hence branching is also essential in every subclass $\mathfrak{C}_0 \subseteq \mathfrak{C}_1$. For removability in $\mathfrak{C}_2$, use the raw residual-conjugacy lift of [3]: every reachable semantic state r has exactly one encoded reachable configuration $\zeta(r)$ in the conjugate graph. Composing the residual coordinate with the bijection $r \mapsto \mu(r)$ gives singleton reachable semantic fibers. □

Proposition 12.5 (Current status of the $A \rightsquigarrow B$ obstruction)**.** *For the literal* 2:1 *presentation,*

$$\boxed{\Pi(A \rightsquigarrow B) = (1, ?, 0).}$$

The obstruction is invariant in its coordinate-isomorphism class and removable after sufficiently broad re-coordination, while fixed-policy essentiality remains open.

Theorem 12.6 (Representation-obstruction separation)**.** *Semantic uniqueness does not imply universality of physical obstructions. Every theorem asserting that a physical branching, debt, schedule conflict, or causal dependency is forced must specify the representation class in which the necessity is claimed.*

Proof. The canonical semantic quotient is unique by Theorem 3.4, while Theorems 11.4 and 12.4 exhibit physical properties that are forced in one representation class but removable in another exact lift of the same semantics. □

13 What is proved, what is scoped, and what remains open

Result	Status	Scope
Right-residual classification	proved	all coprime p, q
Universal semantic quotient	proved	correct deterministic real-time recognizers / input-boundary configs
WSC core $\cong$ residual core	proved	WSC minimum-demand semantics
Canonical-section closure dichotomy	proved	fixed WSC literal orientation / one-top real time
Essential semantic-fiber branching	proved	same fixed literal policy
Coordinate-reduced eager/lazy lifts	proved	stated coordinate morphism class
Exact six-schedule classification	proved	selected 2:1 three-site family
Non-lattice schedule poset	proved	same family
Terminal normalization equivalence	proved	same family / fixed $\mathfrak{N}$
Three-channel canonicality signature	proved	fixed literal policy
Carry visibility and portability criteria	proved	phase-coherent WSC / zero-backlog / one-top settings
Forced K-excess at induced B contexts	proved	2:1 expanded $R0^m$ family
Directed primary dependency $A \to B$	proved	selected 2:1 family
Causal hypergraph coordinate invariance	proved	site-preserving coordinate isomorphism
Semantic non-determination of causal structure	proved	literal-vs-raw coordinate comparison
Branching persistence profile 110	proved	relative to the nested classes defined in Theorem 12.1
Fixed-policy profile of $A \to B$	open	middle coordinate in $(1, ?, 0)$
General downward closure of schedules	open	arbitrary schedule systems
General $p{:}q$ causal hypergraph classification	open	beyond selected 2:1 sites
Classical confluence / Church–Rosser theory	open	not claimed here
Unrestricted DPDA lift minimality	open	not claimed here

14 Executable draft consistency check

The unchanged regression script, `uniqueness_lift_validation_v0_2.py`, reconstructs the 2:1 solid core together with the three optional schedule bits. It is not used as a substitute for the symbolic proofs. As a regression check, it exhausts all binary words of length at most 14 for all eight schedules, verifies the residual invariant on every reached configuration, verifies terminal normalization for all six admissible schedules, and confirms that the first recognition failure of both forbidden schedules 110 and 111 is

$$111110001 \rightsquigarrow (C, 110).$$

The current run performs 458,776 assertion-level checks and passes.

15 Conclusion

For $L_{p,q}$, the right-residual semantics are rigid: every correct deterministic real-time recognizer factors onto the same integer translation system, and the WSC minimum-completion dynamics are an isomorphic semantic coordinate system. Physical pushdown realizations are not rigid. Under the fixed canonical literal policy, one-top locality prevents transition closure for every nontrivial ratio, forcing noncanonical semantic fibers. Even within the stated coordinate morphism category, distinct coordinate-reduced lifts and distinct admissible normalization schedules remain.

The exact 2:1 schedule family makes this nonuniqueness concrete. Six schedules are admissible, two are maximal, the schedule poset is not a lattice, and all six nevertheless normalize to the same canonical endpoint. Their unique minimal conflict is not merely combinatorial: enabling A expands the first-hit observation fiber of B, exposing hidden contexts with different semantic carry under the same local PDA observation. The resulting direct-canonicality failure, carry aliasing, and forced kernel excess are three descriptions of one physical interaction. This causal relation is invariant under coordinate-isomorphic presentations but disappears under a broader residual re-coordination.

Thus the analysis separates at least three nested levels of persistence: presentation-level, fixed-policy, and broader-realization-class persistence. The correct general principle is therefore not that a pushdown obstruction is simply present or absent, but that its persistence must be measured relative to the admissible representation category. This distinction provides the bridge from family-specific WSC normalization to a later general theory of real-time pushdown normalization.

Acknowledgment of companion results

The semantic arithmetic, canonical minimum-demand map, WSC defect coordinates, and literal top-rewrite obstruction framework used here are developed in the frozen WSC manuscript [1]. The centered $p{:}q$ construction and eager/lazy comparison are developed in [2]. The raw residual-conjugacy realization used as the re-coordination witness is developed in [3]. Results imported from those manuscripts are used as foundations rather than re-claimed as new contributions here.

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Code and validation artifacts. Validation scripts and checkpoint material are maintained in the author's public research repositories. The executable checks are supplementary consistency evidence and do not replace the symbolic proofs.

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